

Parth Patel

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Summary: Data-driven Scientist with expertise spanning **cell and gene therapy manufacturing** and **analytical chemistry** in pharmaceutical and biotech settings. Proven experience in mammalian cell culture, TIL/CAR-T therapies, aseptic processing, and bioprocess workflow optimization, coupled with strong analytical skills in **HPLC, GC, LC-MS, ICP-MS/OES, UV-Vis, FTIR, particle size, and dissolution testing**. Adept at **formulation, method development, validation, stability testing, impurity characterization, and regulatory compliance (cGMP, GLP, FDA/EMA)**. Skilled in integrating theoretical knowledge and practical insights to generate high-quality data, troubleshoot complex analytical challenges, and support formulation development and process improvement initiatives.

Technical Skills

Cell Biology & In Vitro Systems: Mammalian cell culture, primary cells, recombinant cell lines, aseptic technique, sterile processing, cell expansion

ADME / Pharmacokinetics / Transporter Science (Relevant Exposure): Drug metabolism (ADME), transporter biology, drug-drug interactions (DDI), cellular uptake/efflux concepts, in vitro assay systems

Bioanalytical & Laboratory Techniques: HPLC, UV-Vis spectroscopy, Rheometer, LC-MS, FTIR, ELISA, ligand binding assays, plate-based assays, flow cytometry, fluorescence microscopy, particle size analyzer, Nanodrop, microplate reader (Epoch), Zetasizer, controlled rate freezer, LOVO system, M1-filler, G-Rex bioreactors.

Data Analysis & Software: GraphPad Prism, STARLIMS, SAPB1, Endnote, ImageJ, OriginPro, Biorender, Inkscape, Microsoft Excel, OriginPro, ImageJ, statistical data analysis, data visualization, trend analysis

Assay Development & Experimental Design: Design of Experiments (DoE), Quality by Design (QbD), assay optimization, method development, troubleshooting

Regulatory & Compliance: GLP, cGMP, GDP, FDA 21 CFR Part 11, batch record documentation, data integrity, audit readiness

Experience

Minaris Advanced therapies

April 2024- present

Cell Therapy Scientist

Philadelphia

Executed tumor dissection, cell expansion, harvest **mammalian cell culture and in vitro biological processes along with** pre-formulation, formulation (**thaw, expansion, harvest**) supporting development of cell therapies (TIL, CAR-T, ceMSCs cell therapies)

- Performed in-process sampling and **bioanalytical testing** (cell count, viability, culture performance), generating high-quality, reproducible data for process monitoring and scientific evaluation
- Developed and optimized HPLC-based analytical workflows to monitor critical small molecules, residual reagents, and metabolites in cell culture media, ensuring batch consistency, reagent quality, and compliance with cGMP standards.
- Applied principles of analytical method development for assay validation, including quantification of cytokines, nutrients, and residual components, supporting regulatory filings and quality assurance processes.
- Supported process development and experimental studies using Design of Experiments (DoE) to evaluate critical process parameters (CPPs), aligning with principles used in in vitro assay development and optimization
- Utilized **data analysis and visualization tools (GraphPad Prism, Excel)** to identify trends, evaluate performance, and support **data-driven decision making**
- Leveraged strong theoretical knowledge of ICH Q3D elemental impurity guidelines and ICP-MS/ICP-OES principles to support compliance strategies and mitigate potential nonconformance risks.

- Spearheaded the scaling up of the team's clinical process to full commercial scale by executing manufacturing process following **FDA, Annex 1 and CFR Title 21** guideline and performed aseptic complex operations in **ISO5 and ISO7** environments.
- **Directly supported FDA and EMA regulatory inspections**, contributing to audit preparation, documentation review, and ensuring continued compliance with cGMP and quality standards.
- Led the implementation of a data-driven continuous improvement protocol that optimized production workflows, resulting in a **20% reduction** in processing time and a **15% decrease** in deviations and non-conforming events.
- Collects, records, reviews, performs analysis, interprets, identify trends of scientific and process data per Good Document Practices (GDP).
- Trained team members in **aseptic techniques, laboratory operations, and standardized workflows**, improving execution consistency and compliance.

Jefferson Methodist hospital

May 2024 – January 2025

Pharmacy technician

Philadelphia

- Prepared patient specific *Intravenous* (IV) formulations in class III BSC (Glovebox) using aseptic techniques, ensuring precision and compliance.
- Supported pharmaceutical operations with strong attention to accuracy, documentation, and quality standards.
- Troubleshoot automated dispensing systems, demonstrating technical problem-solving and adaptability

Research Experience

Thomas Jefferson University | Master's

2022-2023

- Developed **first-of-its-kind PLGA/chitosan nanoparticles** encapsulating **meclizine**, specifically engineered for targeted delivery against **A549 lung cancer cell lines**. (article published in IF:4.5)
- Performed **analytical testing (HPLC, FTIR, UV)** to evaluate physicochemical properties and stability.
- Applied **QbD and DoE principles** to optimize formulation performance
- Conducted **data analysis and interpretation**, generating scientific reports and conclusions
- Conducted **literature-driven and data-supported evaluation of ADME and transporter-related mechanisms**, including **cellular uptake/efflux processes and drug-drug interaction (DDI) considerations**, to strengthen understanding of in vitro assay design and pharmacokinetics
- Analyzed published datasets using **GraphPad Prism and Excel** to interpret trends relevant to **drug metabolism and transporter-mediated kinetics**
- Performed MTT assay, morphological studies, FTIR, drug release, UV, Hoechst dual staining, nanoparticle internalization study, fluorescence microscopy, wound healing assay, HPLC.

Education

• Thomas Jefferson University
Masters in Pharmaceutical Sciences
USA
GPA: 3.8/4

January 2022 – December 2023
Philadelphia,

• OHMO International Research Fellowship

May 2024

Reviewer/Judge

1. Reviewer:

Phytomedicine Plus (Elsevier)

- Harvard Public Health Review: A peer reviewed journal
- Bentham Science journals
- Total reviewed 20+ research articles

2. **Judge:** Invited as judge for abstract, oral and poster presentation for:

- The OHIO Academy of science, Sigma SI, ABRCMS conference

Achievements/Poster presentations

- 1). Event: CHEM-O-MODEL (Model making event) **VES College, Mumbai, 2019**
 - Achieved recognition with **2nd place** in a Model making Competition, for demonstrating **DNA packing and the mechanism of anti-cancer drugs acting on them.**
- 2). Rasayanam (Poster presentation) **Institute of Chemical Technology, Mumbai, 2018**
 - Achieved recognition with **2nd place** in a Model making Competition, for presenting on topic **"Catalytic Converter"**

Publications

- ✓ **Research Interest score: 121.4, Citations: 170,**
- ✓ **ORCID iD: 0000-0001-7083-4885**
- ✓ **SciHorizon: RID-20211355**
- ✓ **Research Gate: <https://www.researchgate.net/profile/Parth-Patel-179>**

- 1) **Parth Patel**, Nicole Oyinade Shoyele, Sunday Adewale Shoyele, Preparation and evaluation of meclizine-loaded PLGA/chitosan nanoparticles as a potential treatment for non-small cell lung cancer using A549 cells, Journal of Drug Delivery Science and Technology, Volume 107, 2025, 106831, <https://doi.org/10.1016/j.jddst.2025.106831> (IF: 4.5)
- 2) **Parth Patel**, Eye disease drug as a potential cure for COVID-19: one-foot-in-the-door, Eye (NaturePortfolio, London, England), 37(14), 2844–2846. <https://doi.org/10.1038/s41433-023-02449-4>
- 3) **Parth Patel**, A bird's eye view on a therapeutically wonder molecule, Phytomedicine Plus, volume 1, issue 3, 2021, <https://doi.org/10.1016/j.phyplu.2021.100070> (Elsevier) (**Excellence review article award, 2nd prize**)
- 4) Asmita Ghosh, Anusua Sarkar, Pubali Paul, **Parth Patel**, The rise in cases of mucormycosis, candidiasis and aspergillosis amidst COVID-19, Fungal Biology reviews (IF:4.06), Vol 38C, 2021, page no: 67-91
- 5) Parth Patel, Abhishek Bansal, Zinc level in hepatocellular carcinoma: a meta-analysis, Journal of cancer prevention & Research 2021; 12(1) :31-34
- 6) Single authored Book chapter: **Parth Patel**, Overview of 3D printing technology: history, types, application and materials, Additive Manufacturing with Medical Applications (1st ed.). CRC Press. <https://doi.org/10.1201/9781003301066>

ADDITIONAL TECHNICAL EXPOSURE

- Foundational understanding of **transporter-related in vitro studies**, including **cellular uptake and inhibition assay principles**
- Familiarity with **LC-MS–based and radiometric quantitation approaches** for bioanalysis and drug metabolism studies
- Exposure to **assay automation concepts and liquid handling systems** in regulated laboratory environments